

DEVIXO SOLUTIONS
DATA SCIENCE INTERNSHIP PROGRAM

TASK 01 – EXPLORATORY DATA ANALYSIS (EDA)

SUPERMARKET SALES ANALYSIS

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Internship Domain: Data Science
Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn

Academic / Internship Project Report

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1. Executive Summary

This report documents the Exploratory Data Analysis performed for Devixo Solutions Data Science Internship Task 01. The project analyzes supermarket transaction data to discover sales patterns, customer behavior, product performance, relationships among numerical variables, and business opportunities.

The dataset contained 1,503 records before cleaning. 3 duplicate records were detected and removed. The dataset contained 30 missing cells before cleaning and 0 after the cleaning procedures. The final analytical dataset contains 1,500 records.

The cleaned data produced total revenue of 438,348.31, an average transaction value of 292.23, and an average customer rating of 6.98/10. Fourteen visualizations, fifteen business insights, and five business recommendations were prepared.

2. Introduction

Exploratory Data Analysis is the process of examining raw data before advanced modeling. It helps identify the structure of a dataset, data-quality issues, distributions, relationships, unusual observations, and patterns that can support business decisions.

This project follows the EDA workflow specified in the internship task: data import and understanding, data cleaning, feature analysis, correlation analysis, visualization, business insights, and recommendations.

The supermarket dataset is suitable for this purpose because it combines numerical variables such as price, quantity, sales, cost and rating with categorical variables such as branch, customer type, product line and payment method.

3. Objectives

- Import and understand the supermarket sales dataset.
- Inspect records, dimensions, columns and data types.
- Identify and handle missing values and duplicate records.
- Check invalid or inconsistent values.
- Analyze numerical and categorical features.
- Identify important business categories and outliers.
- Perform correlation analysis for numerical features.
- Create at least 12 meaningful visualizations.
- Produce 15 evidence-based business insights.
- Provide 5 practical recommendations connected to the findings.

4. Dataset Description

The dataset contains transaction-level supermarket sales information. Each record represents a sales transaction and contains an invoice identifier, date, location, customer information, product information, pricing, quantity, discount, tax, sales value, cost, gross income, payment method and customer rating.

Metric	Value
Records before cleaning	1,503
Duplicate records	3
Records after cleaning	1,500

Missing cells before cleaning	30
Missing cells after cleaning	0
Total revenue	438,348.31
Average transaction	292.23
Average rating	6.98/10

5. Technologies Used

- Python – programming language used for the complete analysis.
- Pandas – dataset loading, cleaning, grouping and statistical analysis.
- NumPy – numerical calculations and correlation preparation.
- Matplotlib – charts and time-series visualization.
- Seaborn – statistical plots and heatmap visualization.
- Jupyter Notebook / Google Colab – interactive execution and documentation.

6. Data Import and Understanding

The CSV dataset was loaded using Pandas. The first and last records were inspected to confirm successful import. The dataset shape and column names were checked, followed by data-type inspection and statistical summaries.

The Date column was converted to datetime format. Numerical and categorical variables were identified using Pandas data-type selection. This separation was useful because numerical variables require statistical and correlation analysis, while categorical variables are better examined using frequency counts and grouped comparisons.

Feature Group	Examples
Numerical	Unit_price, Quantity, Discount, Tax_5pct, Total, COGS, Gross_income, Rating
Categorical	Branch, City, Customer_type, Gender, Product_line, Payment
Date	Date
Identifier	Invoice_ID

7. Data Cleaning

Data cleaning was performed before interpretation. Missing values were first identified. Exact duplicate rows were then removed. The Date column was converted to a consistent datetime type.

Missing categorical values in Customer_type and Payment were replaced using the mode. Missing Rating values were replaced using the median. Basic validity checks were also applied to Quantity, Unit_price, Total and Rating.

Cleaning Step	Result / Decision
Missing values	Categorical values filled with mode; Rating filled with median
Duplicates	3 duplicate rows removed
Date	Converted to datetime
Quantity	Non-positive values checked
Unit price	Non-positive values checked
Total	Non-positive values checked
Rating	Checked against valid 0–10 range

After cleaning, the final dataset was verified again for missing values and duplicates. The cleaned dataset was also exported as `cleaned_supermarket_sales.csv` for reproducibility.

8. Feature Analysis

Product-level analysis shows that Electronic accessories generated the highest total revenue, while Fashion accessories generated the lowest. Branch-level analysis shows Branch A as the highest-revenue branch and Branch C as the lowest-revenue branch.

The payment method with the highest total revenue was Ewallet. The customer type generating the highest total revenue was Member.

Product Line	Transactions	Quantity	Revenue	Avg. Sale	Avg. Rating
Electronic accessories	287	1566	90,552.97	315.52	7.02
Home and lifestyle	259	1433	78,598.63	303.47	6.98
Sports and travel	210	1106	69,376.18	330.36	6.89
Health and beauty	233	1344	68,184.53	292.64	7.06
Food and beverages	278	1541	66,020.39	237.48	6.94
Fashion accessories	233	1320	65,615.61	281.61	6.97

9. Correlation Analysis

Correlation analysis was applied to the numerical variables. Positive correlation means that two variables tend to increase together, while negative correlation indicates an inverse linear relationship. Correlation does not by itself prove causation.

The sales-related variables Total, COGS and Gross_income show important relationships because they are directly connected to transaction value and cost structure. Unit price and quantity are also important because they contribute to the sales amount.

Feature	Unit_price	Quantity	Discount	Tax_5pct	Total	COGS	Gross_income	Rating
Unit_price	1.00	-0.05	0.00	0.63	0.63	0.62	0.55	-0.02
Quantity	-0.05	1.00	-0.04	0.66	0.66	0.65	0.60	0.00
Discount	0.00	-0.04	1.00	-0.11	-0.11	-0.02	-0.28	0.00
Tax_5pct	0.63	0.66	-0.11	1.00	1.00	0.98	0.91	-0.01
Total	0.63	0.66	-0.11	1.00	1.00	0.98	0.91	-0.01
COGS	0.62	0.65	-0.02	0.98	0.98	1.00	0.81	-0.01
Gross_income	0.55	0.60	-0.28	0.91	0.91	0.81	1.00	-0.02
Rating	-0.02	0.00	0.00	-0.01	-0.01	-0.01	-0.02	1.00

10. Data Visualization

The assignment requires meaningful visualization rather than graphs created only to reach a numerical target. The project therefore includes fourteen visualizations covering distribution, categorical comparison, time trends, relationships, outliers and correlations.

- Histogram: Distribution of total sales.

- Bar Chart: Revenue comparison across product lines.
- Count Plot: Transaction counts by customer type.
- Pie Chart: Payment method distribution.
- Scatter Plot: Relationship between unit price and total sales.
- Line Chart: Monthly sales trend.
- Box Plot: Sales distribution and outliers by product line.
- Violin Plot: Sales distribution by customer type.
- Heatmap: Correlation among numerical features.
- Pair Plot: Pairwise relationships among key numerical variables.
- Distribution Plot: Customer rating distribution.
- Area Chart: Monthly sales contribution by product line.
- Branch Sales Chart: Revenue comparison between branches.
- Product Quantity Chart: Quantity sold by product line.

11. Business Insights

- 1. Electronic accessories generated the highest total revenue among product lines.
- 2. Fashion accessories generated the lowest total revenue among product lines.
- 3. Branch A recorded the highest total revenue.
- 4. Branch C recorded the lowest total revenue.
- 5. Ewallet generated the highest revenue among payment methods.
- 6. Member customers generated the highest total revenue among customer types.
- 7. Sports and travel had the highest average transaction value.
- 8. Food and beverages had the lowest average transaction value.
- 9. Health and beauty had the highest average customer rating.
- 10. Electronic accessories had the highest quantity sold.
- 11. Sports and travel had the lowest quantity sold.
- 12. 2025-09 was the strongest month by total revenue.
- 13. 2025-04 was the weakest month by total revenue.
- 14. The overall average transaction value was 292.23.
- 15. The overall average customer rating was 6.98 out of 10.

12. Business Recommendations

- 1. Increase promotional and inventory attention for Electronic accessories, based on its highest revenue contribution.
- 2. Review the pricing, promotion and product mix of Fashion accessories because it recorded the lowest revenue.
- 3. Compare the practices of Branch A with Branch C to identify operational differences.
- 4. Maintain strong support for Ewallet while retaining alternative payment options for customer flexibility.
- 5. Use monthly sales patterns to plan inventory and marketing activity, particularly around 2025-09 and 2025-04.

13. Project Outputs

- Complete Jupyter Notebook containing the EDA code.
- Supermarket sales CSV dataset.
- Cleaned dataset CSV.
- Product-line, branch, customer-type and payment summary files.
- Screenshots of data import, cleaning, analysis and visualizations.
- Professional PDF report.
- README.md file.
- GitHub repository containing the complete project.

14. Conclusion

This project completed the required Exploratory Data Analysis workflow on 1,500 cleaned supermarket sales records. The process covered data understanding, cleaning, feature analysis, outlier checking, correlation analysis and visualization.

The results demonstrate differences in revenue across products, branches, customer types and payment methods, as well as changes in sales across months. The project produced 14 visualizations, 15 business insights and 5 recommendations.

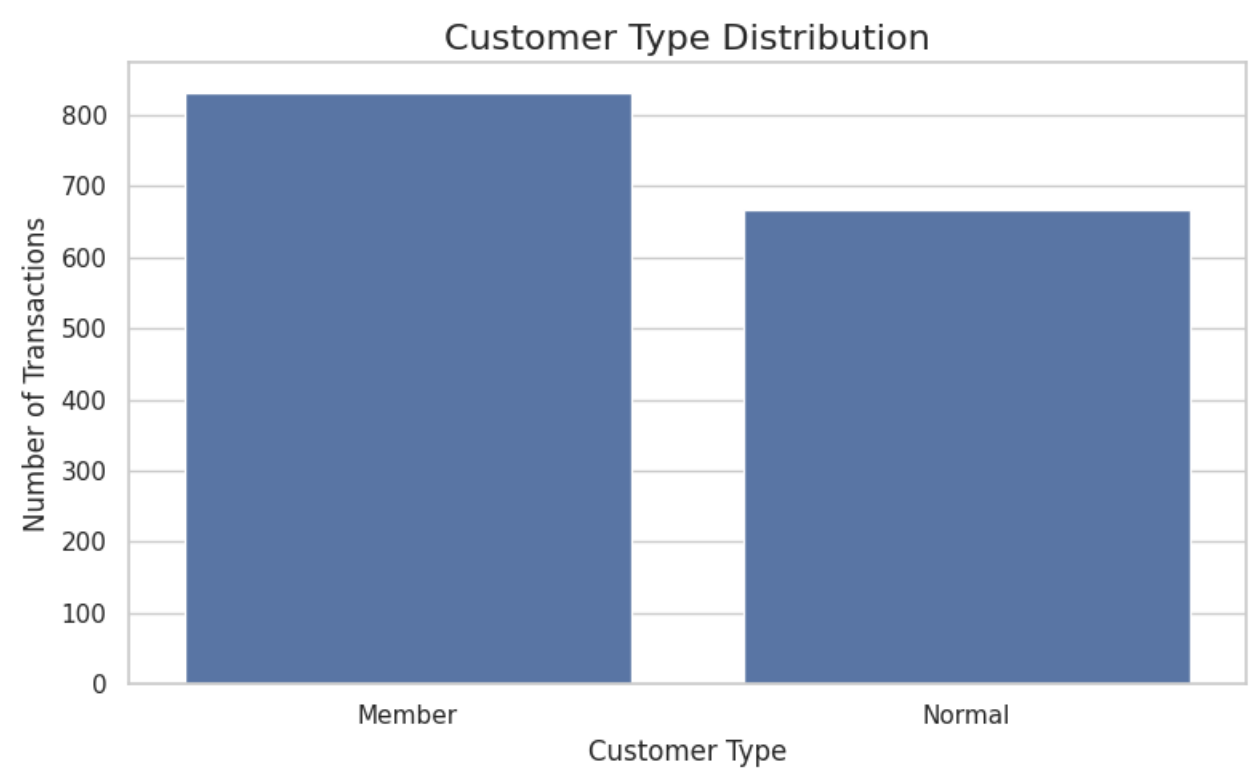
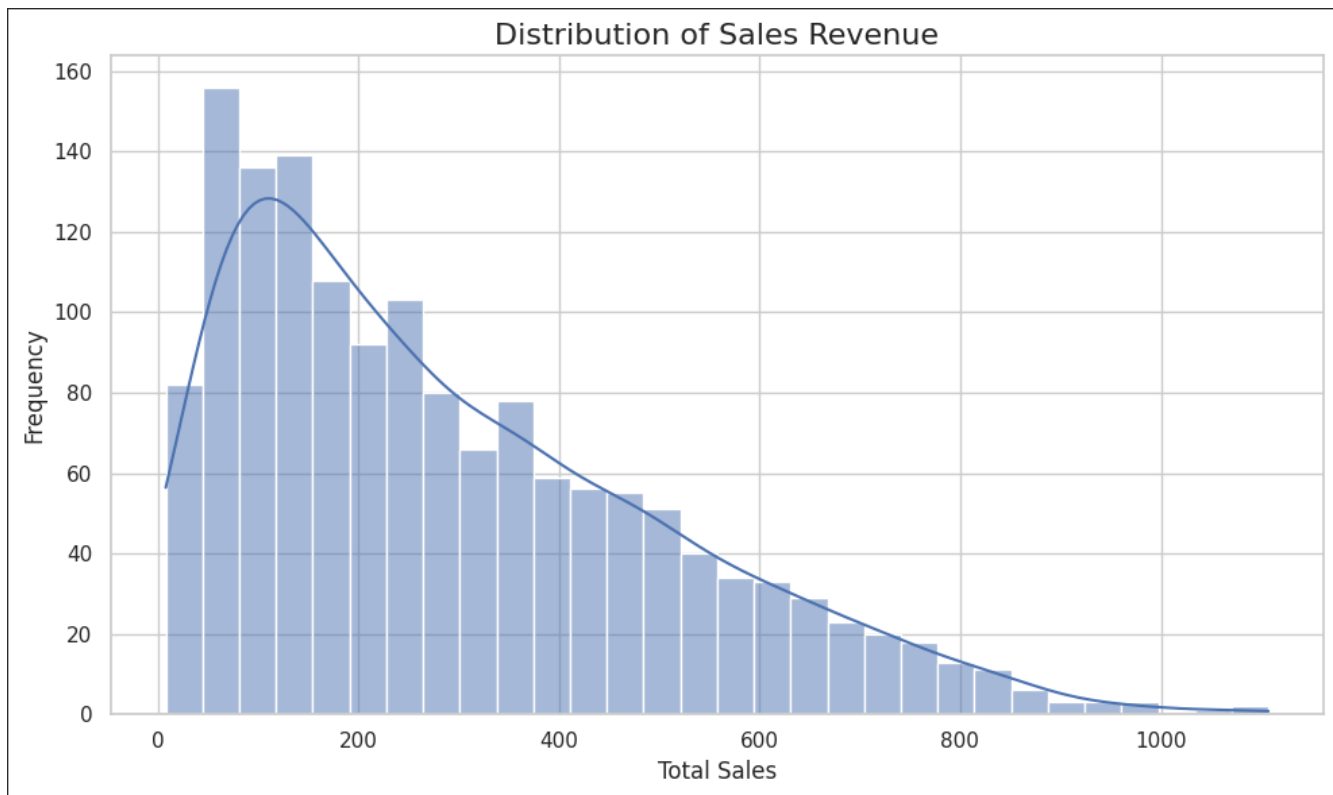
EDA provides a strong foundation for later data science work because it identifies data-quality issues and helps define which features and business patterns deserve further investigation.

15. GitHub Repository Structure

- Devixo-Task-01-EDA/
 - |— Devixo_Task01_Complete_EDA.ipynb
 - |— supermarket_sales_eda_dataset.csv
 - |— cleaned_supermarket_sales.csv
 - |— product_line_summary.csv
 - |— branch_summary.csv
 - |— customer_type_summary.csv
 - |— payment_summary.csv
 - |— screenshots/
 - |— graphs/
 - |— report/
 - | |— Devixo_Task01_EDA_Report.docx
 - |— README.md

References / Task Requirements

Primary source: DEVIXO SOLUTIONS – DATA SCIENCE INTERNSHIP PROGRAM – TASK 01 – EXPLORATORY DATA ANALYSIS (EDA), the assignment document supplied with this project. The task specifies a minimum of 1,000 records, data cleaning, feature analysis, correlation analysis, 12+ meaningful visualizations, 15 business insights, 5 recommendations, screenshots, project documentation and a mandatory GitHub repository.



Payment Method Distribution

